Technical Appendix

One Shot Is All You Need. Or Is It?

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GitHub repository · Hugging Face collection

This appendix records the exact protocols, counts, per-task/per-seed results, training curves, compute measurements, and integrity checks behind the four-page main report. It is not part of the four-page submission limit. All target tables use environment success and show raw counts.

1 Frozen protocol and provenance

1.1 Immutable sources

Item	Frozen value
Base model	`lerobot/smolvla_base`, revision c83c3163b8ca9b7e67c509fffd9121e66cb96205
Dataset	`nvidia/LIBERO_LeRobot_v3`, revision e5907374380b8f96511957e6ba5582be52a1e179
Seen suite	Full available `libero_90`: 3,921 episodes, 569,249 frames, 73 instruction rows
Seen checkpoint	`step_100000`; `weights.pt` SHA-256 2cd510a594a87580f7368b782ca9b37332c0e5002d807093c759e95fbfb57c88
Seen statistics	`libero_90`; digest b159b6fed3e52edf25bd39b377dd64940221b7a030362daf7f726b1c2ecb30cf
Target tasks	Drawer, Bowl, and Wine selected by exact instruction text
First $N = 1$ episodes	Drawer 20; Bowl 13; Wine 6
Train seeds	42 and 123
Target eval	seeds 1000–1019; 20 rollouts per checkpoint; target task statistics
Retention eval	probes `black_bowl_plate`, `drawer_bowl`, `book_caddy`; seeds 1000–1009; `libero_90` statistics
Environment	hard reset, relative control, 300-step horizon, execute 10 of each 50-step action chunk

1.2 Training settings

The vision encoder and VLM backbone stay frozen. Naive training updates the Action Expert and the state/action projections (99,880,992 parameters). Target-LoRA and Replay-LoRA use rank $r = 64$, $\alpha = 64$, zero dropout in the Action Expert, plus trainable projections (4,215,632 parameters).

Stage	LR	Max steps	Stop rule	Batch	Precision
Seen expert	10^{-4}	100,000	fixed	32	bf16
Naive target	10^{-4}	12,000	min(100 epochs, cap)	32	bf16
Target-LoRA	10^{-3}	12,000	min(100 epochs, cap)	32	bf16
Replay-LoRA	10^{-3}	12,000	min(100 epochs, cap)	32	bf16

All runs use AdamW with weight decay 10^{-2}, betas $(0.9, 0.95)$, gradient clipping at 1.0, a 1,000-step warm-up, and cosine decay. Replay-LoRA draws 75% target samples and 25% samples from the full `libero_90` source. No target-success early stopping or rerun is used.

1.3 Gripper and normalization safeguards

The NVIDIA conversion stores gripper action 1 as open, while the environment uses +1 as close. The evaluator applies $a_{\text{env}} = 1 - 2a_{\text{data}}$. Replaying a stored demonstration before training gives success 1.0.

The target policy is deployed with its target statistics. Seen retention is different: it always combines adapted weights with the original `libero_90` statistics. This isolates weight change from statistics change.

2 Complete target and retention counts

2.1 Naive target cost curve

N	Drawer 42	Drawer 123	Bowl 42	Bowl 123	Wine 42	Wine 123	Pooled
0	1/20		0/20		0/20		1/60
1	17/20	19/20	20/20	20/20	16/20	17/20	109/120
2	10/20	20/20	20/20	18/20	16/20	16/20	100/120
5	17/20	14/20	20/20	20/20	17/20	19/20	107/120
10	19/20	20/20	20/20	20/20	19/20	18/20	116/120
25	20/20	20/20	20/20	20/20	16/20	18/20	114/120

N	Success	Rate	95% Wilson interval	Change from $N = 0$
0	1/60	1.7%	[0.3, 8.9]%	–
1	109/120	90.8%	[84.3, 94.8]%	+89.2 pp
2	100/120	83.3%	[75.7, 88.9]%	+81.7 pp
5	107/120	89.2%	[82.3, 93.6]%	+87.5 pp
10	116/120	96.7%	[91.7, 98.7]%	+95.0 pp
25	114/120	95.0%	[89.5, 97.7]%	+93.3 pp

2.2 Corrected seen retention

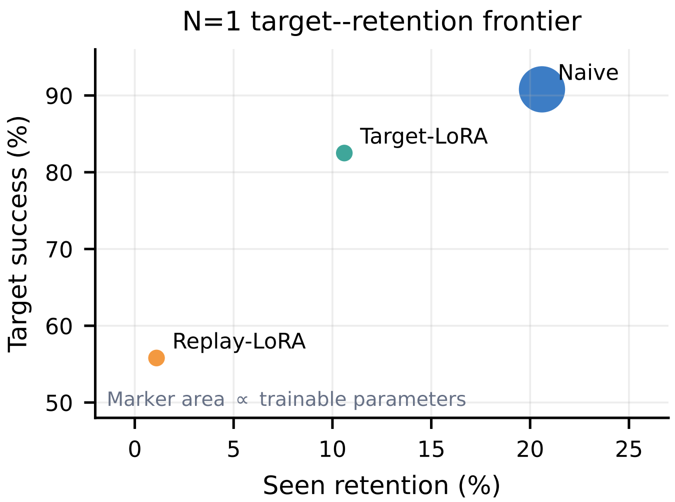
Each adapted row pools two train seeds and three probes, for 60 retention rollouts per target task. The final column pools all six checkpoints at a budget.

N	Drawer-trained	Bowl-trained	Wine-trained	Pooled	Change vs. frozen
0	Frozen reference 24/30 (80.0%)			24/30	–
1	6/60	20/60	11/60	37/180 (20.6%)	-59.4 pp
2	5/60	13/60	3/60	21/180 (11.7%)	-68.3 pp
5	0/60	0/60	0/60	0/180	-80.0 pp
10	0/60	0/60	0/60	0/180	-80.0 pp
25	0/60	0/60	0/60	0/180	-80.0 pp

At the checkpoint level, Pearson correlation between target success and corrected retention is $r = 0.026$. This is descriptive only; the cells are not independent.

2.3 $N = 1$ aggregate comparison

Method	Target	95% CI	Retention	95% CI	Trainable	Wall/cell
Naive	109/120 (90.8%)	[84.3,94.8]	37/180 (20.6%)	[15.3,27.0]	99.9M	162.8 s
Target-LoRA	99/120 (82.5%)	[74.7,88.3]	19/180 (10.6%)	[6.9,15.9]	4.22M	244.0 s
Replay-LoRA	67/120 (55.8%)	[46.9,64.4]	2/180 (1.1%)	[0.3,4.0]	4.22M	309.3 s



3 $N = 1$ per-task and per-seed results

Target is out of 20 rollouts. Retention is out of 30 rollouts (three frozen probes, ten seeds each).

Method	Target task	Seed	Target	Retention	Train wall	Peak VRAM
Naive	Drawer	42	17/20	4/30	199.1 s	7,540 MiB
Naive	Drawer	123	19/20	2/30	199.4 s	7,540 MiB
Naive	Bowl	42	20/20	10/30	162.2 s	7,540 MiB
Naive	Bowl	123	20/20	10/30	162.0 s	7,540 MiB
Naive	Wine	42	16/20	6/30	127.2 s	7,540 MiB
Naive	Wine	123	17/20	5/30	126.9 s	7,540 MiB
Target-LoRA	Drawer	42	15/20	0/30	314.7 s	6,944 MiB
Target-LoRA	Drawer	123	8/20	1/30	315.0 s	6,938 MiB
Target-LoRA	Bowl	42	20/20	9/30	229.7 s	6,944 MiB
Target-LoRA	Bowl	123	20/20	7/30	229.3 s	6,944 MiB
Target-LoRA	Wine	42	19/20	1/30	187.0 s	6,882 MiB
Target-LoRA	Wine	123	17/20	1/30	188.2 s	6,944 MiB
Replay-LoRA	Drawer	42	3/20	0/30	435.9 s	6,944 MiB
Replay-LoRA	Drawer	123	3/20	0/30	436.0 s	6,944 MiB
Replay-LoRA	Bowl	42	20/20	1/30	332.3 s	6,944 MiB
Replay-LoRA	Bowl	123	20/20	1/30	332.3 s	6,944 MiB
Replay-LoRA	Wine	42	8/20	0/30	159.7 s	6,944 MiB
Replay-LoRA	Wine	123	13/20	0/30	159.9 s	6,944 MiB

3.1 Probe totals and instability

Method	Black bowl / plate	Drawer / bowl	Book / caddy
Naive	21/60	14/60	2/60
Target-LoRA	12/60	7/60	0/60
Replay-LoRA	2/60	0/60	0/60

Target-LoRA is stable for Bowl (20/20 for both seeds) but not for Drawer (15/20 vs. 8/20). The result supports adding more train seeds before a final method claim. The three probes should also be expanded only after the method is locked, so probe selection cannot become another form of tuning.

4 Training diagnostics

4.1 Seen expert

The full 100k run takes 38,870 seconds (10 h 47 min 50 s), processes 3.2M samples, reserves 7,536 MiB at the final logged step, and ends at loss 0.23494.

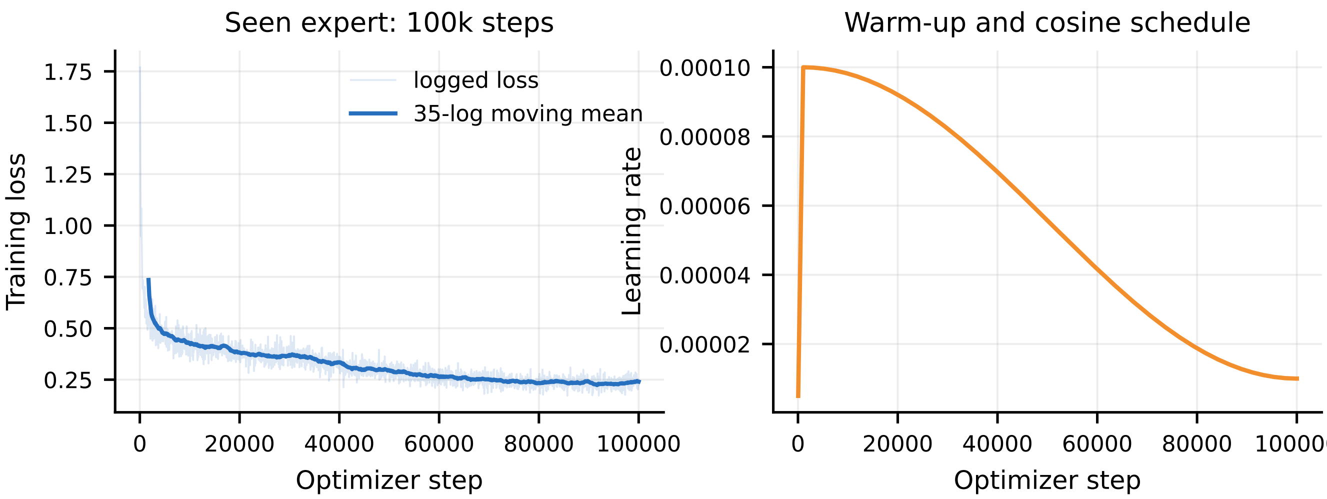


Figure A1. Seen training loss and frozen optimizer schedule.

4.2 Naive target training

The target stop is 100 dataset epochs capped at 12k optimizer steps. Because datasets are small, actual step counts depend on task length and N. For $N = 1$, final logged steps are 300–500. Curves below show the median and interquartile range over three tasks and two train seeds at each budget.

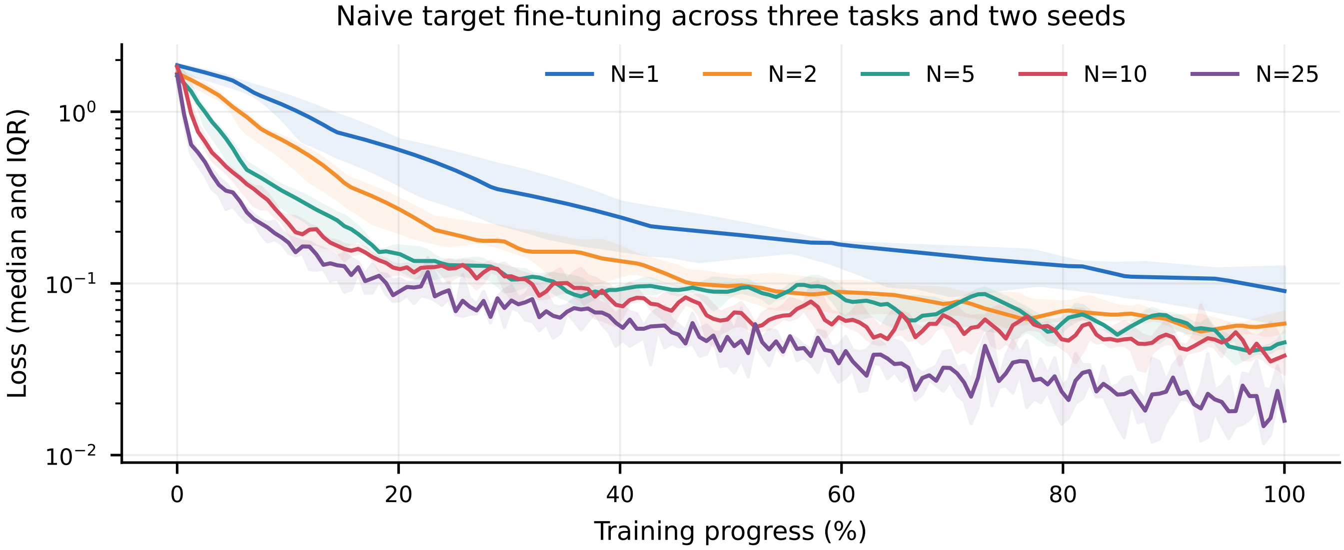


Figure A2. Naive fine-tuning losses. Training progress is normalized because run lengths differ.

4.3 LoRA and replay diagnostics

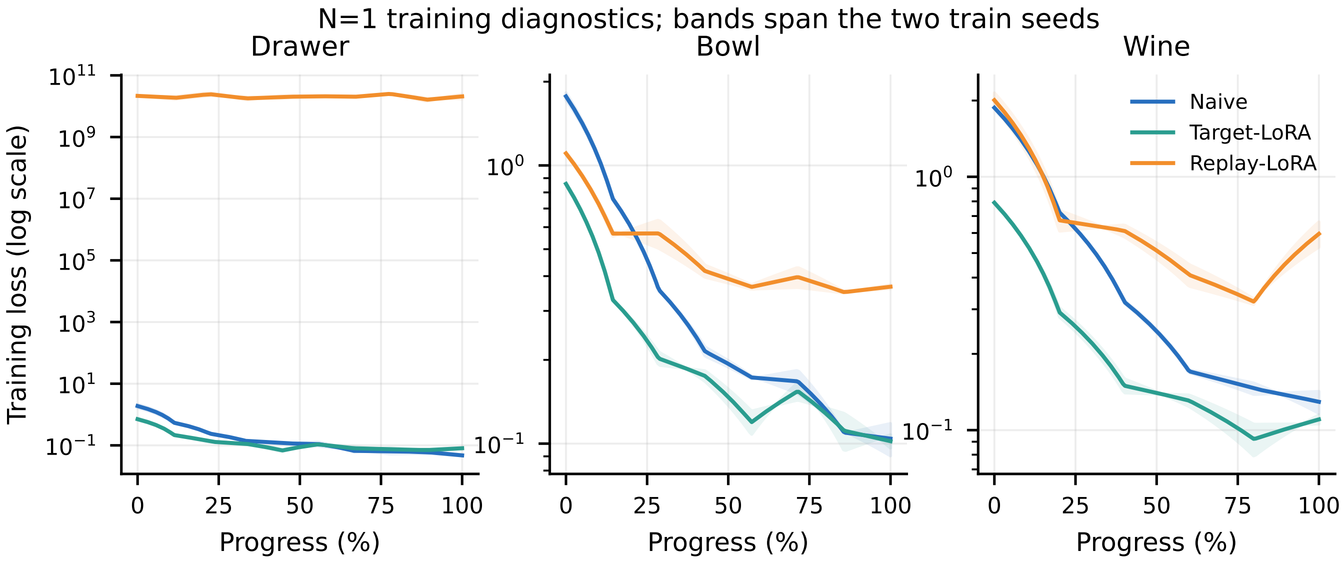


Figure A3. $N = 1$ loss by task and method; bands span seeds 42 and 123.

Replay-LoRA Drawer is numerically pathological. The first Drawer demonstration has gripper standard deviation 10^{-6}, while seen replay contains both gripper states. The target-only normalizer therefore creates extreme standardized replay targets. Final Drawer losses are 1.44×10^{10} and 2.72×10^{10}. Bowl and Wine do not show the same explosion, so this does not explain their full retention loss. A follow-up should log loss and gradient norms by source and by action dimension before any update.

4.4 Compute and concurrency

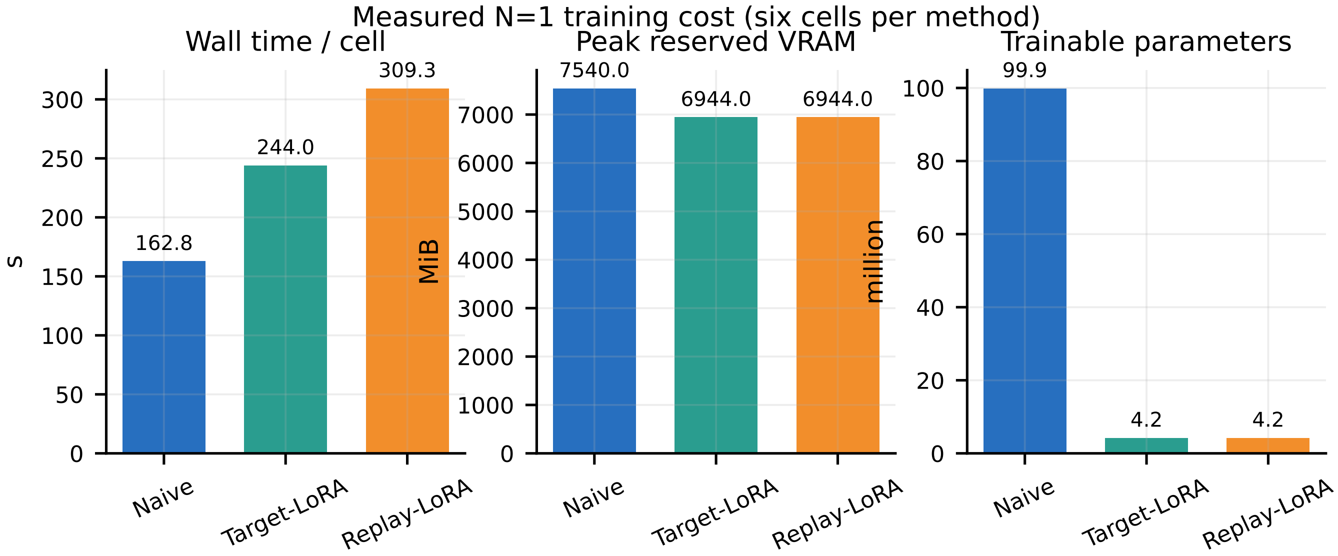


Figure A4. Mean wall time, per-process peak reserved VRAM, and trainable parameter count.

A bounded concurrency benchmark compared two and four simultaneous jobs. Four jobs improved aggregate throughput by about 1.6 times and reserved about 27,776 MiB in total on a 97 GB RTX PRO 6000. The official 12-cell LoRA grid finished in 1,007.7 seconds (16 min 48 s) from first start to last finish. Wall time per cell includes this concurrent load and is not an isolated microbenchmark.

5 Controls, integrity, and artifact map

5.1 Language control

All 60 correct/wrong pairs use identical initial-state fingerprints. The frozen policy changes its trajectory when the instruction changes, but success is too low to prove strong language grounding.

Task	Correct	Wrong	Mean action L_2	Mean cosine gap
Drawer	1/20	0/20	0.5820	0.3273
Bowl	0/20	0/20	0.9982	0.7731
Wine	0/20	0/20	0.9661	0.6626

5.2 Normalization 2-by-2 control

Weights	Statistics	Success	Interpretation
Frozen seen	`libero_90`	12/15	expected seen competence
Frozen seen	target overlay	0/60	statistics alone break deployment
Adapted target	target overlay	0/60	confounded earlier retention
Adapted target	`libero_90`	6/60	corrected weight test

The old 0/900 overlay result is retained as a deployment-normalization warning, not reported as catastrophic forgetting. The corrected retention grid checks checkpoint SHA, statistics digest, task/train/eval seeds, and frozen initial state fingerprints.

5.3 Task 2 integrity

All 12 LoRA training cells are complete and have unique final weight hashes. All 240 target records and 360 retention records passed exact checks for:

- frozen base SHA and method identity;
- task, train seed, evaluation seed, and first-demo episode;
- target or seen statistics suite and digest;
- fixed initial-state fingerprint;
- replay source restricted to `libero_90`, with no selection based on probes.

The code state for the Task 2 grid is clean commit cebe04d9ab408eaa37c7ab0249d48fe915ae7b36. The repository test suite reports 304 passed and 8 skipped.

5.4 Artifact map

Evidence	Location
Main paper source	`report/latex/paper.tex`
Frozen predictions	`predictions.md`
Seen training log	`/mnt/vla/runs/seen__expert__libero90__nall__s42__20260822T010019Z__gd4b8fb8/metrics.csv`
Zero-shot export	`/mnt/vla/eval/zero_shot_v2_seen_stats/report/summary.md`
Language control	`/mnt/vla/eval/language_control/report/summary.md`
Naive $N = 1, 2$ results	`/mnt/vla/validation/TODO28_n12/results.md`
Corrected retention	`/mnt/vla/validation/TODO28_retention_libero90/retention.md`
Task 2 result	`report/tables/task2_n1_results.md`
Task 2 training	`/mnt/vla/runs/task2_n1`
Task 2 target eval	`/mnt/vla/eval/task2_n1/target`
Task 2 retention	`/mnt/vla/eval/task2_n1/retention_libero90`

5.5 Recommended next experiment

Before training, audit one mixed replay batch in the shared action coordinate system and report source counts, per-source/per-dimension normalized ranges, losses, and gradient norms. Then preregister one normalization-safe replay variant or a frozen-policy distillation penalty. Lock the method before evaluating a broader seen-task set, and report the complete target–retention Pareto frontier rather than choosing one checkpoint by target success.